

Title: A Simple 1D Convolutional Neural Network for Signal Classification

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Abstract:

We propose a lightweight 1D Convolutional Neural Network (Conv1DClassifier) for classification of multichannel

Architecture:

The input tensor has shape (batch_size, 2, 1024).

The network consists of the following sequential layers:

1. First Conv1d layer mapping from 2 input channels to 32 output channels with kernel size 7.
2. Batch normalization layer (BatchNorm1d) over 32 features.
3. ReLU activation function.
4. MaxPool1d with kernel size 2.
5. Second Conv1d layer mapping from 32 channels to 64 channels with kernel size 5.
6. Batch normalization layer (BatchNorm1d) over 64 features.
7. ReLU activation function.
8. AdaptiveAvgPool1d with output size 1.
9. Fully connected Linear layer mapping 64 features to 5 output classes.

Requirements:

- Implement the model class Conv1DClassifier in PyTorch.
- Include a model configuration file.
- Provide unit tests to verify the forward pass output shape is (batch_size, 5).